Srijita Bandopadhyay

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RESEARCH SUMMARY

PhD researcher working at the intersection of **Machine Learning**, **Optimization**, **3D Reconstruction** **Inverse Problems**, and **Computational Imaging**, developing data-efficient methods for image restoration.

EDUCATION

Michigan State University East Lansing, MI, USA
Doctor of Philosophy (Ph.D.) in Computational Mathematics, Science, and Engineering *Aug. 2025 – Present*
- **Research Focus**: *Inverse Problems and Deep Learning*

University of Engineering and Management Kolkata, India
Bachelor's in Electronics and Communication Engineering *July. 2019 – July. 2023*
- *GPA*: 9.73/10 (*Rank:1*)
- *Thesis Title*: *Detection of Tomato Leaf Diseases for Agro-Based Industries Using Novel PCA DeepNet*

RESEARCH EXPERIENCE

Graduate Research Assistant, SLIM Lab Aug. 2025 – Present
Michigan State University *East Lansing, MI, USA*
- Developing **3D reconstruction** and image restoration methods that recover clearer images from sparse, noisy, or low-dose measurements in computational imaging.
- Formulating optimization-inspired reconstruction models that combine unrolled algorithms, learned priors, sparsity regularization, and deep image priors.
- Implementing PyTorch pipelines for CT and image restoration experiments, with careful evaluation of artifacts, stability, and reconstruction quality.
- Studying **data-efficient** imaging strategies for settings where clean targets, dense measurements, or paired training data are difficult to obtain.

Undergraduate Research Assistant Jun. 2020 – Apr. 2024
University of Engineering and Management *Kolkata, India*
- Developed deep learning models for plant disease detection, medical image classification, and biomedical image analysis.
- Designed research workflows spanning literature review, model development, training, evaluation, and result visualization.
- Published peer-reviewed research across computer vision, medical imaging, and applied deep learning venues.

PRESENTATIONS

- **Flash Talk:** "Self-supervised High-corruption Image Restoration for Low-dose CT," *CVPR Vision'26*, 2026.
- **Invited Talk:** "AI/ML-Based Smart Systems for Energy, Healthcare and Agricultural Industries," Narula Institute of , 2023.

SELECTED PUBLICATIONS (Google Scholar)

- Haijie Yuan, Chaoyan Huang, **Srijita Bandopadhyay**, Liyue Shen, and Saiprasad Ravishankar, "Fractional-gradient Sparsity with Autoencoding Sequential Deep Image Prior for 3D CT Reconstruction," *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026.
- **Srijita Bandopadhyay**, Joydeep Roy Chowdhury, Rahul Shaw, Swarna Paul and Soumen Banerjee, "Vision Transformer Based Classification of Gastrointestinal Ulcers Using WCE Images", *CIPR-2024* (Link) .
- Bhavaraju Anuraag, **Srijita Bandopadhyay** and Soumen Banerjee, "Lungs Disease Classification Using Transformed Chest X-Ray Images Based on Deep Neural Networks", *CIPR-2024* (Link)
- **Srijita Bandopadhyay**, Kyamelia Roy, Sheli Sinha Chaudhuri, Soumen Banerjee and Korhan Cengiz, "Deep Ensemble Feature Extraction Based Classification of Bleeding Regions Using Wireless Capsule Endoscopy Images", *Intelligent Data Analytics for Bioinformatics and Biomedical Systems, 2024* .

- Kyamelia Roy, Sheli Sinha Chaudhuri, Jaroslav Frnda, **Srijita Bandopadhyay**, Soumen Banerjee and Jan Nedoma, "Diagnosis of Intestinal Diseases Using Endoscopy Images Based on Encoder - Attention DeepNet", *Submitted at* **Signal, Image and Video Processing Journal, Springer** .
- **Srijita Bandopadhyay**, Rahul Shaw, Joydeep Roy Chowdhury, Sanskar Gupta, Kumar Ashish and Soumen Banerjee, "Deep Ensemble Feature Extraction based Classification of Brain Tumor using MRI Images",*IEEE-IATMSI, 2024* (Link) .
- Kyamelia Roy, Sheli Sinha Chaudhuri, **Srijita Bandopadhyay**, IshanJyoti Ray, Yagyashree Acharya, Somava Nath and Soumen Banerjee, "Diagnoses of Covid-19 Using Radiographic Chest X-Ray Images Based on Deep Neural Networks", *In Proceedings of International Conference on Frontiers in Computing and Systems: COMSYS-2022*, vol 690. Springer, 2023.(Link) .

INDUSTRY EXPERIENCE

Cloud Engineer April 2024 – July 2025
LTIMindtree *Mumbai, India*

- Designed and deployed scalable cloud-based systems supporting data-intensive and AI-driven workflows.
- Contributed to development and deployment of machine learning and generative AI pipelines in production environments.

Data Engineering Intern Feb. 2023 – May 2023
LTIMindtree *Kolkata, India*

- **10%** increase in ETL pipeline efficiency by implementing and optimizing workflows using PySpark, Hive, and Shell Scripting.
- Enabled faster insights from Big Data analytics by exploring applications of Machine Learning.

TECHNICAL SKILLS

- **Deep Learning frameworks:** PyTorch, TensorFlow, Keras
- **Programming & Scientific Computing:** Python, MATLAB, NumPy, SciPy, Pandas, LaTeX
- **Machine Learning:** Supervised and unsupervised learning, ensemble methods, regularization techniques
- **Systems & Tools:** Linux, Git, Docker, Jupyter, Anaconda

RELEVANT COURSES

Deep Learning, Machine Learning, Computer Vision, Data Analysis and Visualization, Linear Algebra & Signal Processing

SCHOLASTIC ACHIEVEMENTS

- Chancellor's Award for Academic Excellence for being Department Topper.
- Secured 1st position in the Department of Electronics and Communications for the Academic year [2022-2023].
- Secured 1st position in the Department of Electronics and Communications for the Academic year [2020-2021].

INVITED TALKS

- Faculty Development Program on *"AI/ML-Based Smart Systems for Energy, Healthcare and Agricultural Industries"* organised by the Department of Electronics and Communication Engineering, Narula Institute of Technology.

REVIEWING

- **Computers and Electronics in Agriculture (COMPAG)**, Elsevier, IF = 8.3.